

Folder Structure Reorganization Proposal

Phase 5A — D&D 3.5e Prototype

Date: 2026-05-27
Scope: Assets/Scripts/ (644 C# files across 60+ directories)

1. Executive Summary

The current Assets/Scripts/ hierarchy has grown organically and suffers from **fragmentation**, **mixed concerns**, and a **god-object GameManager** (45 partial files, 44,000+ lines). This proposal reorganizes the codebase into a **domain-driven folder structure** that mirrors game systems, improves discoverability, co-locates tests with their domain, and reduces the cognitive load of finding where any piece of logic lives.

2. Current Structure — Problems Identified

2.1 GameManager God Object (Critical)

Metric	Value
Partial files in Core/	45
Total lines	44,087
Responsibilities	Spell casting, combat actions, NPC setup, loot, treasure, domain powers, walls, grease, mirror image, testing panels, dispel/counterspell

The single GameManager class has accreted every new feature via partial files. This is the single largest architectural debt item.

Files affected:

- GameManager.cs (main)
- GameManager.CombatActions.cs , GameManager.CombatFlowAccessors.cs
- GameManager.SpellCasting.cs , GameManager.DispelCounterspell.cs
- GameManager_Spells_A.cs through GameManager_Spells_W.cs (20 alphabetical spell files)
- GameManager_Spells_Cantrips.cs , GameManager_Spells_Phase1.cs , GameManager_Spells_Phase2.cs , GameManager_Spells_Shared.cs , GameManager_Spells_MagicFang.cs
- GameManager_FlamingSphere.cs , GameManager_Grease.cs , GameManager_WallOfFire.cs , GameManager_WallOfForce.cs , GameManager_WallOfIce.cs
- GameManager_DomainAreaSpells.cs , GameManager_DomainPowers.cs , GameManager_DomainSpells.cs
- GameManager_ConcealmentAreas.cs , GameManager_HolyAreas.cs , GameManager_MirrorImage.cs
- GameManager.NPCSetup.cs , GameManager.NPCTurns.cs

- `GameManager.LootCollection.cs` , `GameManager.TreasureGeneration.cs`
- `GameManager.TestConfigs.cs` , `GameManager.TestPanel.cs`

2.2 Combat Logic Fragmentation

Combat-related code is spread across **4+ top-level folders**:

Location	Files	Contents
Combat/	34	Conditions, calculators, logging, initiative, status effects
CombatSystems/	11	Grapple, overrun, maneuvers, turn undead, melee reactions
Core/GameManager.CombatActions.cs	1	Combat action dispatch
Core/GameManager.CombatFlowAccessors.cs	1	Combat flow accessors
Mounts/	4	Mounted combat system
Services/CombatFlowService.cs	1	Combat flow orchestration

A developer looking for “how attacks work” must search 4+ directories.

2.3 Magic / Spell System Disorganization

Spell logic is scattered across **5 locations**:

Location	Files	Purpose
Core/GameManager_Spells_*.cs	25	Spell resolution by letter (A-W)
Magic/	19	SpellData, SpellDatabase, SpellCaster, casting helpers
Magic/AreaEffects/	31	Area-of-effect spell implementations
Magic/StatusEffects/	28	Buff/debuff data
Magic/Spells/Databases/	23	Spell definition databases (A-Z)
Magic/Components/	7	Concentration, metamagic, summon lists
Magic/Spells/WishExecutor.cs	1	Wish spell
Services/SpellApplicationService.cs	1	Spell application
Services/SpellResolutionService.cs	1	Spell resolution
Services/SpellTargetingService.cs	1	Spell targeting
Services/ConcentrationService.cs	1	Concentration
Services/DispelMagicService.cs	1	Dispel magic

2.4 Equipment / Inventory Overlap

Location	Files	Purpose
Inventory/	13	ItemData, ItemDatabase, potions, scrolls, wands, inventory UI
Equipment/	34	Enchantments, rings, rods, staves, wondrous items, materials
Equipment/Behaviors/	28	Specific magic item behaviors
Store/	2	Store UI and inventory

`ItemData` lives in `Inventory/` but enchantment data lives in `Equipment/`. A “magic sword” requires touching 3+ folders.

2.5 Character Data Bloat

`Character/` contains **65 files** mixing:

- Core stats (`CharacterStats.cs` , `CharacterCombatStats.cs` , `HPState.cs`)
- NPC databases (`NPCDatabase_A.cs` ... `NPCDatabase_Zombies.cs` — 30 files)
- Race data, deity data, domain data, feats, skills
- Templates (`LycanthropeTemplate.cs` , `SkeletonTemplate.cs` , `ZombieTemplate.cs`)
- Unrelated: `RandomEncounterSystem.cs` , `WizardFamiliar.cs`

2.6 Inconsistent Test Organization

Tests are in `Tests/` with per-domain subdirectories, but coverage is uneven:

Test folder	Files
Tests/Combat/	53
Tests/Services/	14
Tests/Character/	10
Tests/Magic/	5
Tests/Classes/	4
Tests/Maneuvers/	4
Tests/Utilities/	3
Tests/Crafting/	1
Tests/Equipment/	1
Tests/Encounters/	1
Tests/Feats/	2
Tests/Inventory/	1
Tests/Mounts/	1
Tests/AI/	1

Tests are already domain-grouped but distant from source. Navigation friction is high.

2.7 Other Issues

- **Systems/** has only 2 files (`EquipmentAssigner.cs` , `QuickSpawnSystem.cs`) — too vague a name for a grab-bag folder.
- **World/** has 2 files — could be merged.
- **Effects/** (Poisons/Diseases) is separate from `Magic/StatusEffects/` and `Combat/` conditions — related but split.
- **Identifiers/** (14 files) — enums and constants that could live with their domains.
- **Data/NPCTemplates/** overlaps with `Character/Templates/` .

3. Proposed New Structure

The reorganization follows **domain-driven grouping**: each game system owns its folder, co-locates its tests, and minimizes cross-folder hunting.

☐	☐	☐	Shared/	
☐	☐	☐	☐ AnimalCompanionData.cs	
☐	☐	☐	☐ Feats/	
☐	☐	☐	☐ Feat.cs	
☐	☐	☐	☐ FeatDefinitions.cs	
☐	☐	☐	☐ FeatManager.cs	
☐	☐	☐	☐ Races/	
☐	☐	☐	☐ RaceData.cs	
☐	☐	☐	☐ RaceDatabase.cs	
☐	☐	☐	☐ Religion/	☐ Domain/deity data
☐	☐	☐	☐ DeityData.cs	
☐	☐	☐	☐ DeityDatabase.cs	
☐	☐	☐	☐ DomainData.cs	
☐	☐	☐	☐ DomainDatabase.cs	
☐	☐	☐	☐ Skills/	
☐	☐	☐	☐ Skill.cs	
☐	☐	☐	☐ ClassSkillDefinitions.cs	
☐	☐	☐	☐ Creatures/	☐ NPC databases, creature types
☐	☐	☐	☐ NPCDatabase.cs	
☐	☐	☐	☐ NPCDatabaseCustom.cs	
☐	☐	☐	☐ NPCDatabase_A.cs	
☐	☐	☐	☐ NPCDatabase_B.cs	
☐	☐	☐	☐ ... (NPCDatabase_C through NPCDatabase_Z)	
☐	☐	☐	☐ NPCDatabase_Dragons.cs	
☐	☐	☐	☐ NPCDatabase_Lycanthropes.cs	
☐	☐	☐	☐ NPCDatabase_Skeletons.cs	
☐	☐	☐	☐ NPCDatabase_Zombies.cs	
☐	☐	☐	☐ DragonData.cs	
☐	☐	☐	☐ SwarmTraits.cs	
☐	☐	☐	☐ CreatureImmunities.cs	
☐	☐	☐	☐ CreatureTypeProgression.cs	
☐	☐	☐	☐ Templates/	☐ Creature templates (merged with Data/NP
☐	☐	☐	CTemplates)	
☐	☐	☐	☐ NPCTemplate.cs	
☐	☐	☐	☐ NPCTemplateDatabase.cs	
☐	☐	☐	☐ TemplateData.cs	
☐	☐	☐	☐ TemplateSpellUpdater.cs	
☐	☐	☐	☐ TemplateSpellValidator.cs	
☐	☐	☐	☐ BaseCreatureDefinitions.cs	
☐	☐	☐	☐ CreatureTemplateFramework.cs	
☐	☐	☐	☐ LycanthropeTemplate.cs	
☐	☐	☐	☐ LycanthropeTemplateBase.cs	
☐	☐	☐	☐ SkeletonTemplate.cs	
☐	☐	☐	☐ SkeletonCreatureTemplate.cs	
☐	☐	☐	☐ ZombieTemplate.cs	
☐	☐	☐	☐ ZombieCreatureTemplate.cs	
☐	☐	☐	☐ CreatureClass/	
☐	☐	☐	☐ CRCalculator.cs	
☐	☐	☐	☐ ClassAssociationRules.cs	
☐	☐	☐	☐ CreatureClassEngine.cs	
☐	☐	☐	☐ ECLTracker.cs	
☐	☐	☐	☐ StatArrayApplier.cs	
☐	☐	☐	☐ Progression/	
☐	☐	☐	☐ LevelUpCalculator.cs	
☐	☐	☐	☐ LevelUpData.cs	
☐	☐	☐	☐ ExperienceCalculator.cs	☐ From Core/
☐	☐	☐	☐ Familiar/	
☐	☐	☐	☐ WizardFamiliar.cs	
☐	☐	☐	☐ Specialization/	
☐	☐	☐	☐ WizardSpecialization.cs	
☐	☐	☐	☐ Combat/	☐ All combat mechanics unified
☐	☐	☐	☐ Core/	☐ Attack resolution, action economy

			AttackCalculator.cs	
			AttackPool.cs	
			ActionEconomy.cs	
			AoOProvokingAction.cs	
			CombatResult.cs	
			FullAttackResult.cs	
			CombatStateMachine.cs	
			DamageModel.cs	
			InitiativeSystem.cs	
			RangeCalculator.cs	
			SizeCategory.cs	
			TeamUtility.cs	
			ThreatSystem.cs	
			Conditions/	Condition data & management
			ConditionManager.cs	
			CombatConditionType.cs	
			AnimateRopeEntangledConditionData.cs	
			AsleepConditionData.cs	
			CharmedConditionData.cs	
			ColorSprayEffectData.cs	
			EnfeebledConditionData.cs	
			FascinatedConditionData.cs	
			FrightenedConditionData.cs	
			TouchOfIdiocyConditionData.cs	
			WebEntangledConditionData.cs	
			Behaviors/	AI-driven condition behaviors
			CharmedBehaviorController.cs	
			ConfusedBehaviorController.cs	
			FascinatedBehaviorController.cs	
			FrightenedBehaviorController.cs	
			StatusEffects/	
			StatusEffect.cs	
			Maneuvers/	From CombatSystems/
			BaseCombatManeuver.cs	
			GrappleSystem.cs	
			OverrunSystem.cs	
			StandardManeuvers.cs	
			SupportActions.cs	
			ICombatSystem.cs	
			Reactions/	Melee reactions
			FireShieldReactionEffect.cs	
			IMeleeReactionEffect.cs	
			MeleeReactionService.cs	
			Special/	Special combat systems
			NegativeLevelSystem.cs	
			TemplateSmiteSystem.cs	
			TurnUndeadSystem.cs	
			Mounts/	From top-level Mounts/
			MountData.cs	
			MountDatabase.cs	
			MountSystem.cs	
			MountedCombatSystem.cs	
			Logging/	
			CombatLogHelper.cs	
			CombatLogger.cs	
			Utilities/	
			CombatCalculationService.cs	
			CombatUtils.cs	
			Tests/	Co-located combat tests
			(53 existing test files from Tests/Combat/)	
			Spell/	Consolidated spell system
			Data/	Spell definitions

			SpellData.cs	
			SpellID.cs	From Identifiers/
			SpellNames.cs	From Identifiers/
			SpellSchool.cs	
			SpellSlot.cs	
			SpellResult.cs	
			SpellRanges.cs	
			SpontaneousCastingData.cs	
			Database/	Spell databases (A-Z)
			SpellDatabase.cs	
			SpellDatabase_A.cs	
			... (SpellDatabase_B through SpellDatabase_Z)	
			SpontaneousCastingType.cs	
			Casting/	Casting pipeline
			SpellCaster.cs	
			SpellCastingHelper.cs	
			SpellComponentSystem.cs	
			SpellSaveResolver.cs	
			SpellUtilities.cs	
			ImbueSpellEntry.cs	
			ImbueWithSpellAbilityManager.cs	
			PartialCasterData.cs	
			Resolution/	GameManager spell partials
			standalone resolvers	
			SpellResolver_A.cs	From Core/GameManager_Spells_A.cs
			SpellResolver_B.cs	From Core/GameManager_Spells_B.cs
			SpellResolver_C.cs	
			SpellResolver_Cantrips.cs	
			SpellResolver_D.cs	
			SpellResolver_E.cs	
			SpellResolver_F.cs	
			SpellResolver_G.cs	
			SpellResolver_H.cs	
			SpellResolver_I.cs	
			SpellResolver_K.cs	
			SpellResolver_L.cs	
			SpellResolver_M.cs	
			SpellResolver_MagicFang.cs	
			SpellResolver_N.cs	
			SpellResolver_P.cs	
			SpellResolver_Phase1.cs	
			SpellResolver_Phase2.cs	
			SpellResolver_R.cs	
			SpellResolver_S.cs	
			SpellResolver_Shared.cs	
			SpellResolver_V.cs	
			SpellResolver_W.cs	
			AreaEffects/	From Magic/AreaEffects/
			AreaEffectManager.cs	
			AreaEffectColors.cs	
			PersistentAreaEffect.cs	
			(all 31 area effect files)	
			Wind/	
			WindEffect.cs	
			WindStrength.cs	
			WindWallAreaEffect.cs	
			GustOfWindEffect.cs	
			StatusEffects/	From Magic/StatusEffects/
			(all 28 status effect data files)	
			ActiveSpellEffect.cs	
			Components/	From Magic/Components/
			ConcentrationManager.cs	
			CounterspellData.cs	

☐	☐	☐	MetamagicData.cs	
☐	☐	☐	MetamagicModifier.cs	
☐	☐	☐	SpellcastingComponent.cs	
☐	☐	☐	StatusEffectManager.cs	
☐	☐	☐	SummonMonsterLists.cs	
☐	☐	☐	Domain/	☐ Domain spell logic from GameManager
☐	☐	☐	DomainAreaSpells.cs	☐ From Core/GameMan-
☐	☐	☐	DomainAreaSpells.cs	
☐	☐	☐	DomainPowers.cs	☐ From Core/GameManager_DomainPowers.cs
☐	☐	☐	DomainSpells.cs	☐ From Core/GameManager_DomainSpells.cs
☐	☐	☐	Special/	☐ Named spell implementations from GameM-
☐	☐	☐	ConcealmentAreas.cs	
☐	☐	☐	ConcealmentAreas.cs	☐ From Core/GameMan-
☐	☐	☐	FlamingSphere.cs	☐ From Core/GameManager_FlamingSphere.cs
☐	☐	☐	Grease.cs	☐ From Core/GameManager_Grease.cs
☐	☐	☐	HolyAreas.cs	☐ From Core/GameManager_HolyAreas.cs
☐	☐	☐	MirrorImage.cs	☐ From Core/GameManager_MirrorImage.cs
☐	☐	☐	WallOfFire.cs	☐ From Core/GameManager_WallOfFire.cs
☐	☐	☐	WallOfForce.cs	☐ From Core/GameManager_WallOfForce.cs
☐	☐	☐	WallOfIce.cs	☐ From Core/GameManager_WallOfIce.cs
☐	☐	☐	WishExecutor.cs	☐ From Magic/Spells/
☐	☐	☐	BonusType.cs	
☐	☐	☐	Tests/	☐ Co-located spell tests
☐	☐	☐	(from Tests/Magic/)	
☐	☐	☐	Equipment/	☐ Unified items, inventory, enchantments
☐	☐	☐	Items/	☐ Core item data
☐	☐	☐	ItemData.cs	☐ From Inventory/
☐	☐	☐	ItemDatabase.cs	☐ From Inventory/
☐	☐	☐	ItemID.cs	☐ From Identifiers/
☐	☐	☐	ItemIDs.cs	☐ From Identifiers/
☐	☐	☐	ItemBuilder.cs	
☐	☐	☐	ItemSpellEffect.cs	☐ From Inventory/
☐	☐	☐	ItemMaterial.cs	
☐	☐	☐	Enchantments/	
☐	☐	☐	EnchantmentEffects.cs	
☐	☐	☐	EnchantmentFactory.cs	
☐	☐	☐	EnchantmentProperties.cs	
☐	☐	☐	EnchantmentStats.cs	
☐	☐	☐	EnchantmentType.cs	
☐	☐	☐	ItemEnchantmentData.cs	
☐	☐	☐	ItemMaterialFactory.cs	
☐	☐	☐	Materials/	
☐	☐	☐	MaterialProperties.cs	
☐	☐	☐	Weapons/	☐ Specific weapon behaviors
☐	☐	☐	DwarvenThrowerBehavior.cs	
☐	☐	☐	FrostBrandBehavior.cs	
☐	☐	☐	HolyAvengerBehavior.cs	
☐	☐	☐	JavelinOfLightningBehavior.cs	
☐	☐	☐	LuckBladeBehavior.cs	
☐	☐	☐	MaceOfSmitingBehavior.cs	
☐	☐	☐	MaceOfTerrorBehavior.cs	
☐	☐	☐	NineLivesStealerBehavior.cs	
☐	☐	☐	OathbowBehavior.cs	
☐	☐	☐	RapierOfPuncturingBehavior.cs	
☐	☐	☐	ShatterspikeBehavior.cs	
☐	☐	☐	SlayingArrowBehavior.cs	
☐	☐	☐	SleepArrowBehavior.cs	
☐	☐	☐	SunBladeBehavior.cs	
☐	☐	☐	SwordOfLifeStealingBehavior.cs	
☐	☐	☐	SwordOfSubtletyBehavior.cs	
☐	☐	☐	SwordOfThePlanesBehavior.cs	

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SylvanScimitarBehavior.cs
Armor/
  AbsorbingShieldBehavior.cs
  AnimatedShieldBehavior.cs
  ArmorOfRageBehavior.cs
  BandedMailOfLuckBehavior.cs
  BreastplateOfCommandBehavior.cs
  CastersShieldBehavior.cs
  DemonArmorBehavior.cs
  LionsShieldBehavior.cs
  MithralFullPlateOfSpeedBehavior.cs
  WingedShieldBehavior.cs
Rings/
  RingAbility.cs
  RingActivationManager.cs
  RingChargeManager.cs
  RingDatabase.cs
  RingFactory.cs
  RingUseTracker.cs
  RingNames.cs
  From Identifiers/
Rods/
  RodData.cs
  RodDatabase.cs
  RodFactory.cs
  RodNames.cs
  From Identifiers/
  MetamagicRodActivation.cs
Staves/
  StaffDatabase.cs
  StaffDefinition.cs
  StaffValidator.cs
Wondrous/
  WondrousItemActivation.cs
  WondrousItemDatabase.cs
  WondrousItemFactory.cs
  WondrousItemNames.cs
  From Identifiers/
SpecificItems/
  SpecificItemBehavior.cs
  SpecificItemDatabase.cs
  SpecificMagicItem.cs
  MagicItemLootGenerator.cs
SpellStorage/
  CounterspellManager.cs
  SpellStorageManager.cs
  StoredSpell.cs
Effects/
  RegenerationEffect.cs
Inventory/
  From top-level Inventory/
  Inventory.cs
  InventoryComponent.cs
  CharacterInventory.cs
  From Character/
  PartyStash.cs
  PotionFactory.cs
  ScrollFactory.cs
  ScrollValidator.cs
  WandFactory.cs
  WandValidator.cs
  RopeItemData.cs
Store/
  From top-level Store/
  StoreInventory.cs
  StoreUI.cs
Tests/
  (from Tests/Equipment/ + Tests/Inventory/)

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AI/	AI systems (unchanged, already well-or-
ganized)	
AIBehaviorData.cs	
AIconsumableManager.cs	
AIProfile.cs	
AISpellcastingStrategist.cs	
LastKnownPositionTracker.cs	
NPCTemplateAIConfigurator.cs	
SpellCategoryClassifier.cs	
SpellcasterAIBehaviorData.cs	
Profiles/	(14 profile files, unchanged)
Custom/	
CustomAIExample.cs	
Tests/	
(from Tests/AI/)	
Encounters/	Encounter system (mostly unchanged)
DiceExpression.cs	
DungeonEncounterExamples.cs	
DungeonEncounterSpawner.cs	
DungeonEncounterTable.cs	
DungeonEncounterTableData.cs	
DungeonEncounterTableEntry.cs	
DungeonEncounterTableExamples.cs	
DungeonEncounterTableManager.cs	
EncounterCSVParser.cs	
EncounterDefinition.cs	
EncounterDescriptionParser.cs	
DungeonEncounters.cs	From Core/GameMan-
ager.DungeonEncounters.cs	
RandomEncounterSystem.cs	From Character/
Tests/	
(from Tests/Encounters/)	
Crafting/	Unchanged, already well-organized
(10 files)	
Tests/	
(from Tests/Crafting/)	
TreasureGenerator/	Unchanged
MagicItemGenerator.cs	
TreasureData.cs	
TreasureDice.cs	
TreasureGenerator.cs	
TreasureResult.cs	
TreasureUI.cs	
Effects/	Poisons & Diseases (renamed for clarity)
Disease.cs	
DiseaseDatabase.cs	
Poison.cs	
PoisonDatabase.cs	
PoisonSpecialEffect.cs	
Grid/	Grid system (unchanged)
PathPreview.cs	
SquareCell.cs	
SquareGrid.cs	
SquareGridUtils.cs	
Services/	Cross-cutting services (slimmed)
AIService.cs	
CombatFlowService.cs	

		ConcentrationService.cs	
		ConditionService.cs	
		DiceService.cs	
		DispelMagicService.cs	
		EconomyService.cs	
		EncounterService.cs	
		InputService.cs	
		MovementService.cs	
		SavingThrowResolver.cs	
		SpellApplicationService.cs	
		SpellResolutionService.cs	
		SpellTargetingService.cs	
		SummoningService.cs	
		TurnService.cs	
		UI/	UI layer (semantically grouped)
		Common/	Shared UI utilities
		DraggableWindow.cs	
		ResizableWindow.cs	
		ScrollbarHelper.cs	
		HoverMarker.cs	
		IconLoader.cs	
		IconManager.cs	
		UIFactory.cs	
		UITheme.cs	
		Combat/	Combat HUD panels
		CombatUI.cs	
		ActionButtonPanel.cs	
		InitiativePanel.cs	
		TargetSelectionPanel.cs	
		CombatLogPanel.cs	
		CombatEndXPUI.cs	
		StatusEffectIndicator.cs	
		StatusEffectTooltipUI.cs	
		QuickItemUsePanel.cs	
		StaffSpellSelectionPanel.cs	
		TurnUndeadTargetSelectionPanel.cs	
		SummonedCreatureVisual.cs	
		CharacterCreation/	(unchanged)
		CharacterCreationManager.cs	
		CharacterCreationData.cs	
		CharacterCreationUI.cs	
		DomainSelectionUI.cs	
		FamiliarSelectionUI.cs	
		WizardSpecializationUI.cs	
		CharacterSheet/	
		CharacterSheetUI.cs	
		CharacterHoverTooltipUI.cs	
		CharacterInfoPanel.cs	
		SkillsUIPanel.cs	
		LevelUpUI.cs	
		Spells/	
		SpellPreparationUI.cs	
		SpellSelectionUI.cs	
		SpellStorageUI.cs	
		SpellTestingPanel.cs	
		DisguiseSelfRaceSelector.cs	
		FeatSelectionUI.cs	
		Inventory/	
		InventoryUI.cs	From Inventory/
		PreCombatInventoryUI.cs	
		LootCollectionUI.cs	
		Encounter/	

EncounterSelectionUI.cs	
EncounterPreviewPanel.cs	
DungeonEncounterGeneratorUI.cs	
RandomEncounterGeneratorUI.cs	
PreCombatHubUI.cs	
Crafting/	
CraftingWorkshopUI.cs	
Wish/	
WishUI.cs	
Utilities/	
DebugCommands.cs	Shared utilities & identifiers
SpriteLoader.cs	From Core/
CameraController.cs	From Core/
SummonCommand.cs	From Core/
AbilityScore.cs	From Identifiers/
DamageType.cs	From Identifiers/
SavingThrow.cs	From Identifiers/
IdentifierExtensions.cs	From Identifiers/
(existing Utilities/ files)	
EquipmentAssigner.cs	From Systems/
QuickSpawnSystem.cs	From Systems/
World/	World systems
CreatureTrapSystem.cs	
PlanarTravelSystem.cs	
Tests/	Integration / cross-cutting tests
Maneuvers/	From Tests/Maneuvers/
Utilities/	From Tests/Utilities/
Feats/	From Tests/Feats/

4. Rationale & Design Principles

4.1 Domain-Driven Grouping

Files are grouped by **game domain** (Combat, Spell, Equipment, Character) rather than by technical layer. This matches how developers think about the D&D ruleset: “I need to fix a spell” → go to `Spell/`.

4.2 Co-located Tests

Each domain folder has a `Tests/` subfolder. This keeps test files physically close to the code they test, reducing context switching.

4.3 Slim Core

`_Core/` (underscore prefix for sort-to-top) contains only the bootstrap and global orchestrator. The `GameManager` shrinks from 45 partials to a thin delegation layer as spell resolution logic moves to `Spell/Resolution/`.

4.4 Consolidated Spell System

All 110+ spell-related files converge into `Spell/`. The alphabetical `GameManager_Spells_X.cs` files become `Spell/Resolution/SpellResolver_X.cs` — same content, proper home.

4.5 Unified Equipment

`Inventory/` , `Equipment/` , `Equipment/Behaviors/` , and `Store/` merge into one `Equipment/` tree with semantic subdivisions (Weapons, Armor, Rings, Rods, etc.).

4.6 Character Hierarchy

65 `Character/` files get meaningful sub-folders: Stats, Classes, Feats, Races, Creatures (NPC databases), Templates, Progression.

4.7 UI Semantic Grouping

Flat `UI/` and `UI/Panels/` reorganize into `UI/Combat/` , `UI/CharacterSheet/` , `UI/Spells/` , `UI/Inventory/` , etc.

5. Folders Eliminated

Old Folder	Reason	New Location
<code>CombatSystems/</code>	Merged into Combat	<code>Combat/Maneuvers/</code> , <code>Combat/Reactions/</code> , <code>Combat/Special/</code>
<code>Mounts/</code>	Combat subdomain	<code>Combat/Mounts/</code>
<code>Inventory/</code> (top-level)	Merged into Equipment	<code>Equipment/Inventory/</code> , <code>Equipment/Items/</code>
<code>Store/</code>	Merged into Equipment	<code>Equipment/Store/</code>
<code>Systems/</code>	Only 2 files, vague name	<code>Utilities/</code>
<code>Identifiers/</code>	Distributed to owning domains	<code>Spell/Data/</code> , <code>Equipment/Items/</code> , <code>Equipment/Rings/</code> , <code>_Core/</code> , <code>Utilities/</code>
<code>Data/NPCTemplates/</code>	Merged with Character templates	<code>Character/Templates/</code>
<code>Classes/</code> (top-level)	Merged into Character	<code>Character/Classes/</code>
<code>Magic/</code>	Renamed & restructured	<code>Spell/</code>
<code>Core/</code>	Slim down; most files relocated	<code>_Core/</code> (lean)
<code>UI/Panels/</code>	Distributed to semantic groups	<code>UI/Combat/</code> , <code>UI/CharacterSheet/</code> , etc.

6. Migration Plan

Phase 5B Execution Order

Migration should proceed in dependency order, with one commit per subsystem:

Step	Subsystem	Files Moved	Risk	Priority
1	Combat unification	~50 files	Medium — many cross-references	High
2	Spell consolidation	~110 files	High — GameManager partials	High
3	Equipment merge	~75 files	Medium	High
4	Character reorganization	~65 files	Low — mostly internal	Medium
5	UI semantic grouping	~42 files	Low — UI is leaf layer	Medium
6	Cleanup (Identifiers, Systems, Core slim)	~20 files	Low	Low
7	Test co-location	~100 files	Low — tests only	Low

Migration Strategy

1. **One subsystem per commit** — easy bisect if something breaks
2. **Update namespace declarations** after each move (if applicable)
3. **Run tests after each step** — ensure nothing breaks
4. **Keep `.meta` files** (Unity asset references) — move them alongside `.cs` files
5. **Update any hardcoded paths** in test fixtures or asset references

7. Benefits

Benefit	Before	After
Find spell logic	Search 5+ folders	Go to <code>Spell/</code>
Find combat logic	Search 4+ folders	Go to <code>Combat/</code>
Find item/equipment	Search 3 folders	Go to <code>Equipment/</code>
GameManager complexity	45 partials, 44K lines	Thin orchestrator + domain services
Test proximity	Root <code>Tests/</code> folder	Co-located <code>Tests/</code> in each domain
Onboarding time	High — must learn fragmented layout	Low — intuitive domain folders

8. Risks & Mitigations

Risk	Impact	Mitigation
Broken references after moves	Build errors	Unity handles <code>.meta</code> GUIDs; careful <code>.cs</code> namespace updates
Merge conflicts with active branches	Dev friction	Coordinate with team; do in a single feature branch
GameManager partial class moves	Complex refactor	Phase 2 focuses specifically on this; keep <code>partial class GameManager</code> declarations intact initially
Test breakage	CI fails	Run full test suite after each step
Lost git blame history	Hard to trace changes	Use <code>git mv</code> for all moves to preserve history

9. Implementation Readiness

- ☒ Full directory analysis complete
- ☒ All 644 files cataloged

-  Problem areas identified with file counts
 -  New structure designed with rationale
 -  Migration plan with phased approach
 -  File mapping CSV generated (see `FOLDER_STRUCTURE_MAPPING.csv`)
 -  Ready for Phase 5B execution
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Document generated as part of Phase 5A — Folder Structure Analysis & Reorganization Plan